E 5.1

a) maximize Δ

subject to $x_{12} - x_{23} - x_{24} = 0$

$x_{13} + x_{23} - x_{34} = 0$

$\Delta - x_{12} - x_{13} = 0$

$x_{24} + x_{34} - \Delta = 0$

$0 \le x_{12} \le 8$

$0 \le x_{13} \le 7$

$0 \le x_{23} \le 2$

$0 \le x_{24} \le 4$

$0 \le x_{34} \le 12$

```python
import cvxpy as cp

x12 = cp.Variable()
x24 = cp.Variable()
x23 = cp.Variable()
x34 = cp.Variable()
x13 = cp.Variable()
delt = cp.Variable()

constraints = [
    x12-x23-x24 == 0,
    x13+x23-x34 == 0,
    delt-x12-x13 == 0,
    x24+x34-delt == 0,
    x12>=0,x12<=8,
    x13>=0,x13<=7,
    x23>=0,x23<=2,
    x24>=0,x24<=4,
    x34>=0,x34<=12
]

obj = delt

problem = cp.Problem(cp.Maximize(obj), constraints)
problem.solve()

print(f"x12 = {x12.value:.5f}, x13 = {x13.value:.5f}, x23 = {x23.value:.5f}, x24 = {x24.value:.5f}, x34 = {x34.value:.5f}, optimal_value = {obj.value:.5f}")
```

```
PS C:\Users\lenovo> & C:/Users/lenovo/AppData/Local/Programs/Python/Python313/python.exe c:/Users/lenovo/Desktop/大二下/MAT3007/mat/hw5_q1.py
x12 = 6.00000, x13 = 7.00000, x23 = 2.00000, x24 = 4.00000, x34 = 9.00000, optimal_value = 13.00000
PS C:\Users\lenovo>
```

The optimal solution is:

$x = \begin{bmatrix} x_{12} \\ x_{13} \\ x_{23} \\ x_{24} \\ x_{34} \end{bmatrix} = \begin{bmatrix} 6 \\ 7 \\ 2 \\ 4 \\ 9 \end{bmatrix}$, with optimal value = 13

b) Dual: minimize $8z_{12} + 7z_{13} + 2z_{23} + 4z_{24} + 12z_{34}$

subject to $z_{12} \ge y_1 - y_2$

$z_{13} \ge y_1 - y_3$

$z_{23} \ge y_2 - y_3$

$z_{24} \ge y_2 - y_4$

$z_{34} \ge y_3 - y_4$

$y_1 - y_4 = 1$

$z_{ij} \ge 0$

$\therefore$ The complementarity conditions:

$x_{ij}(z_{ij} - (y_i - y_j)) = 0$, $\forall (i,j)$,

$z_{ij}(c_{ij} - x_{ij}) = 0$, $\forall (i,j)$

That is: $x_{12}(z_{12} - 1 + y_2) = 0$ $\quad z_{12}(8 - x_{12}) = 0$

$x_{13}(z_{13} - 1 + y_3) = 0$ $\quad z_{13}(7 - x_{13}) = 0$

$x_{23}(z_{23} - y_2 + y_3) = 0$ $\quad z_{23}(2 - x_{23}) = 0$

$x_{24}(z_{24} - y_2) = 0$ $\quad z_{24}(4 - x_{24}) = 0$

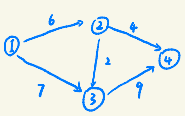
$x_{34}(z_{34} - y_3) = 0$ $\quad z_{34}(12 - x_{34}) = 0$

$\begin{bmatrix} x_{12} \\ x_{13} \\ x_{23} \\ x_{24} \\ x_{34} \end{bmatrix} = \begin{bmatrix} 6 \\ 7 \\ 2 \\ 4 \\ 9 \end{bmatrix}$

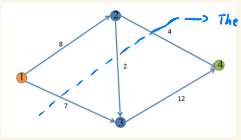
$\therefore$ We can get an optimal solution:

$z = \begin{bmatrix} z_{12} \\ z_{13} \\ z_{23} \\ z_{24} \\ z_{34} \end{bmatrix} = \begin{bmatrix} 0 \\ 1 \\ 1 \\ 1 \\ 0 \end{bmatrix}$, with the optimal value 13

c) The corresponding maximum flow:



minimum cut:



→ The minimum cut.

E 5.2

$$\begin{aligned} \max \quad & 5x_1 + 10x_2 \\ \text{s.t.} \quad & x_1 + 3x_2 \le 50 \\ & 4x_1 + 2x_2 \le 60 \\ & x_1 \le 5 \\ & x_1, x_2 \ge 0 \end{aligned}$$

B	0	0	$\frac{10}{3}$	0	$\frac{5}{3}$	175
2	0	1	$\frac{1}{3}$	0	$-\frac{1}{3}$	15
4	0	0	$-\frac{2}{3}$	1	$-\frac{10}{3}$	10
1	1	0	0	0	1	5

Table 1: Simplex Tableau for Sensitivity Analysis

The original problem → Standard form:

min $-5x_1 - 10x_2$

s.t. $x_1 + 3x_2 + x_3 = 50$

$4x_1 + 2x_2 + x_4 = 60$ $\Rightarrow A = \begin{bmatrix} 1 & 3 & 1 & 0 & 0 \\ 4 & 2 & 0 & 1 & 0 \\ 1 & 0 & 0 & 0 & 1 \end{bmatrix}$

$x_1 + x_5 = 5$,

$x_1, x_2, x_3, x_4, x_5 \ge 0$

$\therefore$ a) The optimal solution is $\begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 5 \\ 15 \end{bmatrix}$, $\begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \\ x_5 \end{bmatrix} = \begin{bmatrix} 5 \\ 15 \\ 0 \\ 10 \\ 0 \end{bmatrix}$

The optimal value of the original problem is 175.

(b) The optimal basis is $\{2, 4, 1\}$,

$A_B = \begin{bmatrix} 3 & 0 & 1 \\ 2 & 1 & 4 \\ 0 & 0 & 1 \end{bmatrix}$ $\quad A_B^{-1} = \begin{bmatrix} \frac{1}{3} & 0 & -\frac{1}{3} \\ -\frac{2}{3} & 1 & -\frac{14}{3} \\ 0 & 0 & 1 \end{bmatrix}$ $\quad \bar{c}_N^T = \begin{bmatrix} \frac{10}{3} & \frac{5}{3} \end{bmatrix}$, $A_B^{-1} A_N = \begin{bmatrix} \frac{1}{3} & -\frac{1}{3} \\ -\frac{2}{3} & -\frac{14}{3} \\ 0 & 1 \end{bmatrix}$

To make the current optimal basis remain optimal, the optimal solution should be still feasible

$\therefore x_B = A_B^{-1}(b + \Delta b) = x_B^* + A_B^{-1}\Delta b = x_B^* + \lambda A_B^{-1} \cdot \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} \ge 0$

$\therefore \begin{bmatrix} 15 \\ 10 \\ 5 \end{bmatrix} + \lambda \begin{bmatrix} \frac{1}{3} & 0 & -\frac{1}{3} \\ -\frac{2}{3} & 1 & -\frac{14}{3} \\ 0 & 0 & 1 \end{bmatrix} \cdot \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} = \begin{bmatrix} 15 + \frac{\lambda}{3} \\ 10 - \frac{2}{3}\lambda \\ 5 \end{bmatrix} \ge 0$ $\quad \therefore -45 \le \lambda \le 15$

$\therefore$ The range of b_1 is $b_1 \in [5, 65]$ so that the current optimal basis still remains optimal

(c) $\because$ For $b_1 = 60 \in [5, 65]$

$\therefore$ The optimal basis is still $\{2, 4, 1\}$

$\therefore x_B = A_B^{-1} b = \begin{bmatrix} \frac{1}{3} & 0 & -\frac{1}{3} \\ -\frac{2}{3} & 1 & -\frac{14}{3} \\ 0 & 0 & 1 \end{bmatrix} \cdot \begin{bmatrix} 60 \\ 60 \\ 5 \end{bmatrix} = \begin{bmatrix} \frac{55}{3} \\ \frac{50}{3} \\ 5 \end{bmatrix}$

$\therefore$ The optimal value is $10 \times \frac{55}{3} + 5 \times 5 = \frac{625}{3}$

$\therefore$ The new optimal primal solution is $\begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 5 \\ \frac{55}{3} \end{bmatrix}$, $\begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \\ x_5 \end{bmatrix} = \begin{bmatrix} \frac{5}{3} \\ \frac{55}{3} \\ 0 \\ \frac{100}{3} \\ 0 \end{bmatrix}$ the new optimal value is $\frac{625}{3}$

(d) $\because$ 2 is in the basis, $2 \in B$

$\therefore$ To make the current optimal basis remain optimal

$\Rightarrow$ The reduced cost $= \bar{c}_N^T - \lambda e_1 \cdot A_B^{-1} A_N \ge 0$

$\begin{bmatrix} \frac{10}{3} & \frac{5}{3} \end{bmatrix} - \lambda \begin{bmatrix} 1 & 0 & 0 \end{bmatrix} \cdot \begin{bmatrix} \frac{1}{3} & -\frac{1}{3} \\ -\frac{2}{3} & -\frac{14}{3} \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} \frac{10}{3} - \frac{\lambda}{3} & \frac{5}{3} + \frac{\lambda}{3} \end{bmatrix} \ge 0$

$\therefore -5 \le \lambda \le 10$

The range of c_2 is $c_2 \in [0, 15]$ so that the current optimal basis still remains optimal